

KARAN NAYAL

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Education

Graphic Era Hill University

Bachelor of Technology in Computer Science and Engineering.

Aug. 2023 - June 2027

Projects

YOLOv8 Object Tracker | *Python, YOLOv8, DeepSORT, OpenCV*

- Built a high-performance computer vision system for real-time object detection, tracking, and counting in CCTV footage using **YOLOv8**.
- Integrated **DeepSORT** to assign and maintain unique IDs for detected objects across consecutive video frames, effectively handling temporary object occlusions.
- Processed video streams using **OpenCV** to render bounding boxes, tracking metrics, and dynamic data overlays for visual analysis.
- Optimized model inference pipelines to ensure high frame rates for smooth, real-time video processing and accurate tracking.

ConnectAlum | *MongoDB, Express.js, React.js, Node.js, JavaScript, CSS*

- Architected and developed a **full-stack** educational networking platform to connect students with alumni, featuring secure role-based access control and custom user dashboards.
- Engineered a real-time bi-directional messaging system and live notification engine to facilitate seamless mentorship requests, appointment scheduling, and direct communication.
- Built an advanced analytics dashboard for alumni to track engagement metrics, manage pending student requests, and publish exclusive job and internship opportunities.
- Implemented real-time data synchronization across the platform, updating engagement charts and notification badges instantly without page reloads to ensure a highly responsive user experience.

Student Print Hub | *Node.js, Express.js, MongoDB, JavaScript, HTML/CSS*

- Built a comprehensive **full-stack** e-commerce platform for automated student document printing and robust order management using **Node.js** and **Express.js**.
- Integrated Google **OAuth 2.0** and the **Razorpay** API to establish secure user identities and process real-time digital payments, effectively streamlining the billing workflow.
- Processed dynamic order lifecycles and document uploads using **MongoDB** for high-performance data querying and **Cloudinary** for secure, scalable cloud storage.
- Engineered a centralized Admin Dashboard (CRM) to track order statuses, monitor payment records, and automate customer communications, ensuring maximum operational efficiency.

Achievements

- Guardian** on LeetCode with a Contest Rating of **2168**.
- Achieved a Best **Global Rank** of 9,042, placing in the **Top 1.08%** among LeetCode Weekly Contest participants.
- Secured Institute **Rank 60** on **GeeksforGeeks** with a Coding Score of 890.
- Ranked 4256 in the **Unstoppable** 100-Day Coding Sprint.
- Campus Mantri — **GeeksforGeeks** : Served as the official campus representative, promoting coding culture, technical initiatives, and student engagement programs.
- Finalist— **IIIT Delhi** Research Tank Hackathon.
- Participated in 5+ hackathons demonstrating rapid prototyping, team collaboration, and innovation under pressure.

Technical Skills

Languages: Python, C++, SQL, JavaScript, R

Full Stack development: HTML, CSS, React.js, Next.js, Express.js, RESTful APIs, JWT Authentication

Databases: MongoDB, MySQL

Data Analytics & Visualization: Power BI, Advanced Excel with AI, Matplotlib

ML & CV: TensorFlow, Scikit-learn, OpenCV

Tools: VS Code, PyCharm, IntelliJ, Git, GitHub, Canva, Microsoft Office, Docker, Jupyter Notebook, Linux

Coding Profiles

- LeetCode:** Solved 441+ Data Structures & Algorithms problems across diverse topics, demonstrating strong problem-solving, algorithmic thinking, and coding proficiency.
- GeeksForGeeks** : Solved 270+ Data Structures & Algorithms problems, reinforcing core programming concepts through consistent coding practice.

Relevant Coursework

- Data Structures and Algorithms
- Object-Oriented Programming
- Operating Systems
- Database Management Systems
- Computer Networks

Certification

- SQL Mastery for Data Science – GeeksforGeeks, 2025
- Power BI– Data Visualization– GeeksforGeeks, 2026
- AWS Training and Certification– Amazon Web Services, 2025
- Data Analytics Virtual Experience– Accenture (Forage), 2024
- Mastering Advanced System Design and Architecture – GeeksforGeeks, 2026